1. Clinical Question & Core Hypothesis

Clinical Question

From the Ontologist synthesis, high TMB or dMMR/MSI-H should sensitize MIBC to immune checkpoint blockade, yet many TMB-high tumors fail to respond; the puzzle is what tumor- or host-level features block the expected immunologic benefit. 2) Tier 1 paths used: I built on the TMB core path (mechanistic link: neoantigen load → potential ICI benefit) and the liquid/tissue tumor biology axis anchored by TP53 (frequently mutated in MIBC and linked to aggressive, immune-evasive phenotypes). I selected these because they provide a genomic (TMB) and tumor-intrinsic (TP53) axis that can interact with microenvironmental imaging features. 3) Tier 2 evidence that supports the claim: SARIFA and stromal metrics (microenvironment/invasion-front patterns) and HDL-C–linked systemic metabolic signals were flagged as modifiers that can alter immune infiltration/function; DPYD and PTGDS indicate metabolic/drug-response heterogeneity that could further modulate clinical effect. 4) Why this endpoint: I chose therapy_response (ICI response proxy) because the core mechanistic question is why genomic predictors of immunogenicity (TMB/dMMR) do not uniformly translate into ICI benefit — a therapy-response endpoint directly tests that discrepancy and is actionable clinically. 5) What I excluded and why: I deliberately excluded MSI/dMMR as a co-anchor to keep the hypothesis focused (TMB already captures mutational load) and omitted exploratory/less-established paths (BCG trial biomarkers, GARS1) from the core claim to preserve mechanistic clarity; these are retained as secondary_modifiers for follow-up analyses?

Working Hypothesis

We hypothesize that in muscle-invasive bladder cancer (MIBC), a high tumor mutational burden (TMB) and TP53-mutant status create a baseline propensity for neoantigen-driven T cell responses, but this pro-immunogenic effect is functionally overridden by a tumor microenvironment with a high stromal-to-tumor ratio (TSR). A high TSR—reflecting dense extracellular matrix, cancer-associated fibroblast activity, and physical immunosuppressive barriers—restricts neoantigen presentation and CD8+ T cell infiltration/effector function, thereby dampening response to immune checkpoint blockade (ICI) even in TMB-high tumors. This interaction between genomic drivers (TMB, TP53) and microenvironmental architecture (TSR) yields a distinct resistance phenotype that can be detected clinically on pre-treatment pathology and imaging and stratifies patients for combination therapies targeting stroma alongside ICI. The hypothesis implies testable predictions: (1) among TMB-high MIBC, TSR will be inversely associated with ICI response, and (2) TP53-mutant, TSR-high tumors will show the weakest response compared with TSR-low counterparts, while TSR-low, TMB-high TP53-mutant tumors will have comparatively better responses. Secondary modifiers (MSI/dMMR status, SARIFA patterns, HDL-C–associated metabolism, DPYD/PTGDS signatures, and circulating tumor cells) may modulate the magnitude of this effect but are not required for the core mechanism. Experimental approaches will test this axis across clinical cohorts, spatial biology, and mechanistic systems. Feasibly, TSR can be quantified from standard H&E or multiplex IHC using digital pathology; the proposed work will integrate TMB (and selective MSI/dMMR assessment), TP53 status, and TSR with immune contexture to predict ICI response and guide rational combination strategies (e.g., stromal/CAF targeting or TGF-β pathway inhibition alongside PD-1/PD-L1 blockade). This framework yields testable hypotheses and actionable clinical implications for stratified therapy in MIBC.

Element	Value
Imaging feature	Stromal-to-tumor ratio (TSR)
Anchor gene(s)	TMB, TP53
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Primary hypothesis gene(s)

TMB: TMB already captures mutational load) and omitted exploratory/less-established paths (BCG trial biomarkers, GARS1) from the core claim to preserve mechanistic clarity; these are retained as secondary_modifiers for follow-up analyses.

TP53: TP53 (frequently mutated in MIBC and linked to aggressive, immune-evasive phenotypes).

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	MSI/dMMR	Supporting modifier from knowledge graph
Secondary	SARIFA	Supporting modifier from knowledge graph
Secondary	HDL-C	Supporting modifier from knowledge graph
Secondary	DPYD	Supporting modifier from knowledge graph
Secondary	PTGDS	Supporting modifier from knowledge graph
Secondary	CTCs	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: TMB and neoantigen load correlate with ICI responsiveness but do not guarantee benefit in MIBC, indicating modulators beyond neoantigen quantity influence outcomes;

Prediction 2: TP53 mutations are common in MIBC and associate with genomic instability and immune-evasive tumor phenotypes, potentially dampening immunogenicity despite high TMB;

Prediction 3: the tumor stroma and ECM (captured by TSR) can physically impede immune infiltration and foster immunosuppressive signaling, a phenomenon supported by literature on SARIFA, stromal remodeling, and CAF-driven TGF-β signaling. By integrating TSR with TMB and TP53 status, we can identify a microenvironment-mediated resistance phenotype and develop strategies to overcome this barrier.

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Novel mechanistic hypothesis derived from knowledge graph analysis.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- MSI/dMMR: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- SARIFA: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.